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Final Report

The Art of Scientific Computation: Electron Wavefunction Approximations and Computational Methods

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1 Introduction

1.1 Abstract

Electron orbital behaviour within both atomic and cyclic-molecular systems was investigated using approximate Hamiltonians of varying complexities. When compared with experimental results [13][10], the Hückel method [16] produced accurate electronic wavefunctions for both cyclic-planar Benzene molecules and periodic Graphene crystals. The Hartree-Fock algorithm [6][12] produced accurate s-type Hydrogen energies in both analytical and numerical forms, but required computationally-expensive resolutions for accurate fully-numerical orbitals. An error was found in the algorithm used for the Hartree-Fock potential $G^{(HF)}$, and as a result, molecular Hartree-Fock results for both Benzene and Graphene could not be compared to those from the Hückel method. Therefore, of the methods successfully tested, the Hückel method most accurately described the complex orbitals within Graphene.

1.2 Motivations

Understanding the behaviour of electrons within atomic and molecular systems has been essential towards designing and manufacturing a large proportion of the electronic devices we use and rely on today [8]. Additionally, having accurate and efficient models of these behaviours allows us to more easily understand them, and develop more advanced tools using them. One particularly interesting group of these quantum systems is that of Graphene crystals, which have a variety of useful electronic properties for devices, including high electron mobilities and lattice stability, and accessible chemical doping [7]. Therefore, by comparing the qualities of our existing models of Graphene electron orbitals, we can interpret their relative utilities.

1.3 Aim and Expectations

The aim of this study is to analyse the behaviour and accuracy of two orbital electron models within atomic and cyclic-planar molecular potentials, particularly in the periodic structure of Graphene. This will be achieved computationally using the Hückel [16] and Hartree-Fock [12] algorithms, which approximately model cyclic-planar and general molecular potentials, respectively. We expect to see calculated atomic energies closely aligning with well-known values, such as for ground-state Hydrogen [11] and Helium [9], but predict that molecular results may diverge from experimental values by varying degrees, due to the simplifying assumptions used by each algorithm.

2 Background

2.1 Electron Wavefunctions

In quantum mechanics, the behaviour of particles within a static potential $\hat{V}(\boldsymbol{r})$ can be modelled with the time-independent Schrödinger equation, an operator eigenvalue equation that acts on the energy-dependent quantum states $|\psi\rangle$ of the particles with the Hamiltonian operator $\hat{H}$, to find the total energy eigenvalue E of each state [8]:

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle \tag{2.1}$$

By applying an orthonormal basis of position states $|\boldsymbol{r}\rangle$ to this equation, it can be rewritten in terms of the inner product of these states instead, their position-dependent state wavefunctions: $\psi(\boldsymbol{r}) = \langle \boldsymbol{r}|\psi\rangle$ [8]. Then, the Schrödinger equation for an electron in a static potential has the following form, in terms of the kinetic and potential energy operators, $\hat{K}$ and $\hat{V}(\boldsymbol{r})$ [12]:

$$\hat{H}\,\psi(\boldsymbol{r}) = \hat{K}\,\psi(\boldsymbol{r}) + \hat{V}(\boldsymbol{r})\,\psi(\boldsymbol{r}) = (E_k + E_V)\,\psi(\boldsymbol{r}) = E\,\psi(\boldsymbol{r}) \tag{2.2}$$

2.2 Basis States and Eigenvalue Equations

Just as quantum states can be described with the weighted sum of basis states $|\boldsymbol{r}\rangle$, particle wavefunctions can be written as the weighted sum of a complete set of position-dependent basis functions, $f(\boldsymbol{r})$ [12]. Then, with the choice of an accurate basis for each wavefunction, they can be approximated with the sum of a finite, incomplete set of N basis functions. Using this, we can describe the wavefunction of an atomic electron with the following equation, weighted by the state-specific expansion coefficients, c_{ai} [12]:

$$\Psi_k(\boldsymbol{r}) \approx \sum_{i}^{N} c_{ki} \cdot f_i(\boldsymbol{r}) \tag{2.3}$$

One accurate basis for atomic electron orbitals is the Slater basis, of functions $f_i(\boldsymbol{r_i})$ [12]. These functions have the following form, where l_i and m_i are the orbital angular momentum (azimuthal) and magnetic angular momentum quantum numbers of the atom, λ_i are real-valued parameters, and $Y_{l_i}^{m_i}(\theta_i, \phi_i)$ are the spherical-harmonic basis functions [4]:

$$f_i(\boldsymbol{r_i}) = N_i r_i^{l_i} \exp[-\lambda_i r_i] Y_{l_i}^{m_i}(\theta_i, \phi_i), \quad \text{with} \quad N_i = \sqrt{\frac{(2\lambda_i)^{l_i+3/2}}{(2l_i+2)!}} \tag{2.4}$$

Another commonly chosen basis is the Gaussian basis, which has the following, similar form. Here, λ_i are a basis of n parameters such as the geometric basis, with $\lambda_i = \alpha\beta^{i-1}$ for some α and β, and $\Gamma(z)$ is the mathematical Gamma function. This basis is less accurate to the shapes of electron orbitals, but has a significant advantage in the Gaussian product theorem [2], which allows the Gaussian components $\exp[-\lambda_i r_i^2]$ of these functions to be multiplied together to produce another Gaussian, significantly simplifying integration of their products [12].

$$f_i(\boldsymbol{r_i}) = N_i r_i^{l_i} \exp[-\lambda_i r_i^2] Y_{l_i}^{m_i}(\theta_i, \phi_i), \quad \text{with} \quad N_i = \sqrt{\frac{2(2\lambda_i)^{l_i+3/2}}{\Gamma(l_i + 3/2)}} \tag{2.5}$$

Finally, with the electron wavefunctions represented in terms of these basis functions, we can separate the Schrödinger eigenvalue equation into smaller components, and instead solve for the expansion coefficients $\vec{c}_k$ using the following generalised eigenvalue equation, where $\hat{S}$ and $\hat{F}$ are the degrees of spatial overlap and interaction between each pair of basis functions respectively, and ϵ_k is the kth energy eigenvalue [12]:

$$\hat{F}\, c_k = \epsilon_k \hat{S}\, c_k \tag{2.6}$$

These can be solved numerically using the finite-difference method, where the operators and basis functions are replaced by finite-sized matrices $\boldsymbol{F}$ and $\boldsymbol{S}$, and vectors $\vec{c}_k$ respectively. There are various approximations available to the electron Hamiltonian for different atomic and molecular systems, and each have corresponding interaction and overlap matrices $\boldsymbol{F}$ and $\boldsymbol{S}$. We will explore some of these in this project.

2.3 Hückel Approximation

One such approximation is the Hückel molecular orbital approximation, which models the behaviour of the π-electrons of cyclic-planar compounds using an identity $\boldsymbol{S}$ matrix, and a $\boldsymbol{F}$ matrix made of two parameters, α and β [12]. This is a simple approximation, and is only valid for the quantum numbers $l = 1$, $m = 0$, but can give accurate results for these limited states [16]. The Hückel matrix elements are selected using the nearest atomic neighbours of each atom in the modelled compound. The α parameters lie on the diagonals of the matrix, and represent atomic interactions, while the β parameters are placed at (i, j) indices where the ith and j^{th} atoms are neighbours, and interact more strongly. Therefore, as an example, the Hückel matrix for a ring-shaped molecule has the following structure [12]:

$$\boldsymbol{F}^{(ring-shaped)} = \begin{pmatrix} \alpha & \beta & 0 & \dots & 0 & \beta \\ \beta & \alpha & \beta & \dots & 0 & 0 \\ 0 & \beta & \alpha & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \alpha & \beta \\ \beta & 0 & 0 & \dots & \beta & \alpha \end{pmatrix} \tag{2.7}$$

2.4 Hartree-Fock Approximation

Another, more in-depth approximation is the Hartree-Fock method, used for general atomic and molecular systems. Here, the interaction matrix $\boldsymbol{F}$ is modelled as the sum of the single-particle kinetic energy $\boldsymbol{T}$, the nuclear Coulomb potential $\boldsymbol{V}$, and a self-consistent electron self-interaction (or Hartree-Fock) potential, $\boldsymbol{G}^{(HF)}$ [6]:

$$\boldsymbol{F} = \boldsymbol{T} + \boldsymbol{V} + \boldsymbol{G}^{(HF)} \tag{2.8}$$

In this approximation, the overlap and kinetic energy matrices $\boldsymbol{S}$ and $\boldsymbol{T}$ have the following index forms, as integrals over the basis functions $f_i(\boldsymbol{r})$ [12]:

$$S_{ij} = \int f_i^*(\boldsymbol{r}) f_j(\boldsymbol{r})\, d\boldsymbol{r} \tag{2.9}$$

$$T_{ij} = -\frac{\hbar^2}{2m} \int f_i^*(\boldsymbol{r}) \nabla^2 f_j(\boldsymbol{r})\, d\boldsymbol{r} \tag{2.10}$$

Then, the nuclear potential matrix $\boldsymbol{V}$ has the following index form, as a sum over distance-weighted atomic charges, within another integral over the basis functions $f_i(\boldsymbol{r})$ [12]:

$$V_{ij} = \frac{e^2}{4\pi\epsilon_0} \int f_i^*(\boldsymbol{r}) \left(\sum_{I=1}^{M} -\frac{Z_I}{|\boldsymbol{r} - \boldsymbol{R}_I|} \right) f_j(\boldsymbol{r})\, d\boldsymbol{r} \tag{2.11}$$

Finally, if the system modelled has more than one electron, the Hartree-Fock potential $\boldsymbol{G}^{(HF)}$ has the following index form, as the sum over the $\boldsymbol{J}$ and $\boldsymbol{K}$ inverse-distance matrices, multiplied by the expansion-coefficient density matrix $\boldsymbol{D}$. Note that the $\boldsymbol{J}$ and $\boldsymbol{K}$ matrices are equal under the index swap ij, kl to il, kj [12]:

$$G_{ij}^{(HF)} = \sum_{k=1}^{N} \sum_{l=1}^{N} (2J_{ij,kl} - K_{ij,kl})\, D_{kl} \tag{2.12}$$

$$D_{kl} = 2 \sum_{a=1}^{N} c_{ka}^* \cdot c_{la} \tag{2.13}$$

$$J_{ij,kl} = \frac{e^2}{4\pi\epsilon_0} \iint f_i^*(\boldsymbol{r}) f_j(\boldsymbol{r}) \frac{1}{|\boldsymbol{r} - \boldsymbol{s}|} f_k^*(\boldsymbol{s}) f_l(\boldsymbol{s})\, d\boldsymbol{r}\, d\boldsymbol{s}, \quad \textbf{and} \quad K_{ij,kl} = J_{il,kj} \tag{2.14}$$

Conveniently, by using Gaussian basis functions $f_i(\boldsymbol{r_i})$ and applying a combination of the Gaussian product theorem and the multi-centre Boys' integrals, these operator matrix integrals can be analytically evaluated [12]. For single-atom systems, this results in the following significantly simplified forms for each Hartree-Fock matrix:

$$S_{ij}^{(atomic)} = \frac{(4\lambda_i\lambda_j)^{3/4}}{(\lambda_i + \lambda_j)^{3/2}} \tag{2.15}$$

$$T_{ij}^{(atomic)} = \frac{3}{4} \frac{(4\lambda_i\lambda_j)^{7/4}}{(\lambda_i + \lambda_j)^{5/2}} \tag{2.16}$$

$$V_{ij}^{(atomic)} = -\frac{\sqrt{32}Z}{\sqrt{\pi}} \frac{(\lambda_i\lambda_j)^{3/4}}{(\lambda_i + \lambda_j)} \tag{2.17}$$

$$J_{ij,kl}^{(atomic)} = \frac{16}{\sqrt{\pi}} \frac{(\lambda_i\lambda_j\lambda_k\lambda_l)^{3/4}}{(\lambda_i + \lambda_j)(\lambda_k + \lambda_l)\sqrt{\lambda_{ijkl}}} \tag{2.18}$$

Then, using the same method, the integrals for the $\boldsymbol{S}$, $\boldsymbol{T}$ and $\boldsymbol{V}$ Hartree-Fock matrices reduce to the following forms in molecular systems, for each atomic combination, AB. Note that $\boldsymbol{A}$ and $\boldsymbol{B}$ here are the 3D positions of each atom in the AB pair [12]:

$$S_{ij}^{AB\,(molecular)} = \frac{(4\lambda_i\lambda_j)^{3/4}}{(\lambda_i+\lambda_j)^{3/2}} R_{ij}^{AB}, \quad \textbf{with} \quad R_{ij}^{AB} = \exp\left[-\frac{\lambda_i\lambda_j}{\lambda_i+\lambda_j}|\boldsymbol{A}-\boldsymbol{B}|^2\right] \tag{2.19}$$

$$T_{ij}^{AB\,(molecular)} = \frac{3(4\lambda_i\lambda_j)^{7/4}}{4(\lambda_i+\lambda_j)^{5/2}} R_{ij}^{AB}\left[1-\frac{2\lambda_i\lambda_j}{3(\lambda_i+\lambda_j))}|\boldsymbol{A}-\boldsymbol{B}|^2\right] \tag{2.20}$$

$$V_{ij}^{AB\,(molecular)} = -\frac{(4\lambda_i\lambda_j)^{3/4}}{(\lambda_i+\lambda_j)} R_{ij}^{AB}\sum_{I=1}^{M} Z_I \frac{\text{erf}\left[\sqrt{v_{il,kj}^{AB,I}}\right]}{\sqrt{v_{ij}^{AB,I}}}, \tag{2.21}$$

$$\textbf{with} \quad v_{ij}^{AB,I} = (\lambda_i+\lambda_j)|\boldsymbol{P}^{AB}-\boldsymbol{R}_I|^2, \quad \textbf{and} \quad \boldsymbol{P}^{AB} = \frac{\lambda_i\boldsymbol{A}+\lambda_j\boldsymbol{B}}{\lambda_i+\lambda_j}$$

Finally, the $\boldsymbol{J}$ and $\boldsymbol{K}$ matrices have the following forms (through index swaps), contributing to a Hartree-Fock potential matrix $\boldsymbol{G^{(HF)}}$ for each atomic combination, AB [12]:

$$J_{ij,kl}^{AB\,(molecular)} = 8\sum_{C=1}^{M}\sum_{D=1}^{M} R_{ij}^{AB}R_{kl}^{CD}\frac{(\lambda_i\lambda_j\lambda_k\lambda_l)^{3/4}}{(\lambda_i+\lambda_j)(\lambda_k+\lambda_l)\sqrt{\lambda_{ijkl}}}\frac{\text{erf}\left[\sqrt{v_{il,kj}^{AB,CD}}\right]}{\sqrt{v_{ij,kl}^{AB,CD}}}, \tag{2.22}$$

$$\textbf{with} \quad v_{ij,kl}^{AB,CD} = \frac{(\lambda_i+\lambda_j)(\lambda_k+\lambda_l)}{\lambda_{ijkl}}|\boldsymbol{P}^{AB}-\boldsymbol{Q}^{CD}|^2$$

For each of these, the $v_{ij}^{AB,I}$ and $v_{ij,kl}^{AB,CD}$ matrices have all values equal to zero for the atomic combinations $A = B$, and $A = B = C = D$, respectively. In these cases, to avoid inaccurate calculations in the $\boldsymbol{V}$, $\boldsymbol{J}$ and $\boldsymbol{K}$ matrices, we use the following limit [12]:

$$\lim_{x\to 0}\left(\frac{\text{erf}[\sqrt{x}]}{\sqrt{x}}\right) = \frac{2}{\sqrt{\pi}}$$

Together, each of these equations allows us to construct both the atomic and molecular Hartree-Fock methods as efficient numerical algorithms. These approximations and their numerical implementations will be further explored in the rest of this report.

3 Method

The Hückel and Hartree-Fock approximations were both implemented in Python using the NumPy, SciPy, and Matplotlib libraries. These methods were separated into general functions, that were used to generate atomic positions and basis functions, and specific functions used to generate and solve the Hamiltonian matrices for each method. Atomic units were used in these scripts to simplify calculations. Refer to Table 3.1 below for the atomic-unit constants and conversion factors used [1].

Description		**Value (Atomic)**		**Value (SI)**
Electronic charge	e	1	or	$1.60218 \cdot 10^{-19}$ C
Proton mass	m_p	$1.83615 \cdot 10^{3}$	or	$1.67262 \cdot 10^{-27}$ kg
Neutron mass	m_n	$1.83835 \cdot 10^{3}$	or	$1.67493 \cdot 10^{-27}$ kg
Electron mass	m_e	1	or	$9.10938 \cdot 10^{-31}$ kg
Permittivity of vacuum	ϵ_0	$1/4\pi$	or	$8.85419 \cdot 10^{-12}$ F/m
Speed of light in vacuum	c	$1.37036 \cdot 10^{2}$	or	$2.99792 \cdot 10^{8}$ m/s
Planck constant	h	2π	or	$6.62607 \cdot 10^{-34}$ J $\cdot$ s
Reduced Planck constant	$\hbar$	1	or	$1.0546 \cdot 10^{-34}$ J $\cdot$ s
Electron volt	eV	$3.67493 \cdot 10^{-2}$	or	$1.60218 \cdot 10^{-34}$ J
Bohr radius	a_0	1	or	$5.29177 \cdot 10^{-11}$ m
Angstrom	$\mathring{A}$	1.88973 a_0	or	$1 \cdot 10^{-10}$ m

Table 3.1: A table of the relevant constants and conversion factors used in this experiment, in both atomic and SI units. The results in this report were calculated using atomic units, which both simplified manual calculations and accelerated project code execution.

3.1 Molecule Construction

The atomic positions corresponding to Benzene and Graphene molecules were both calculated in different ways within the project script. For Benzene, molecules were created by first choosing a central point, and then distributing atoms around this point at equal radii but evenly spaced azimuthal angles. For the Hückel method, only Carbon atoms were distributed in this way, but for the Hartree-Fock approximation, both Carbon and Hydrogen atoms were distributed at equal angles but different radii. Here, the radii chosen were $R_C = R_{(CC)} = 1.3913$ $\mathring{A}$, and $R_H = R_{(CC)} + R_{(CH)} = 2.4722$ $\mathring{A}$, corresponding to existing experimental values [5]. Refer to Figure 3.1 below for the resulting atomic positions used for Benzene.

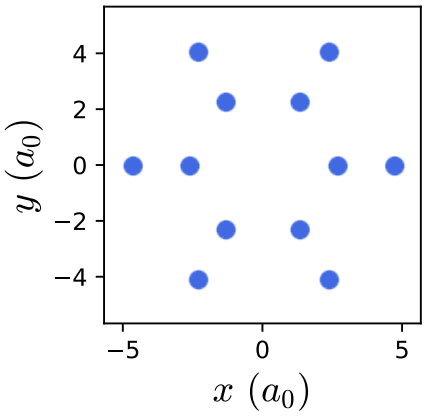


Figure 3.1: Molecular structure of the Benzene molecule considered in this report. The Carbon atoms are separated by 1.3913 angstrom, or approximately 2.63 Bohr radii (a_0). The Hydrogen atoms are then 1.0809 angstrom, or approximately 2.04 Bohr radii (a_0) from the nearest Carbon atom.

For the hexagonal lattice of Graphene, a more complex method was used. The Carbon atoms were first arranged in rows, and were then offset horizontally to create a triangular lattice. After this, the script iterated through lattice positions, and removed atoms that lay at hexagonal centre-points, creating the desired hexagonal lattice. Here, the lattice points were spaced at distances of $R_{CC} = 1.42\ \mathring{A}$, corresponding to their experimental separation [15]. Refer to Figure 3.2 below for the three Graphene lattices used in this report, G_1, G_2, and G_3, which were created using initial row sizes of 5, 7, and 10 atoms respectively.

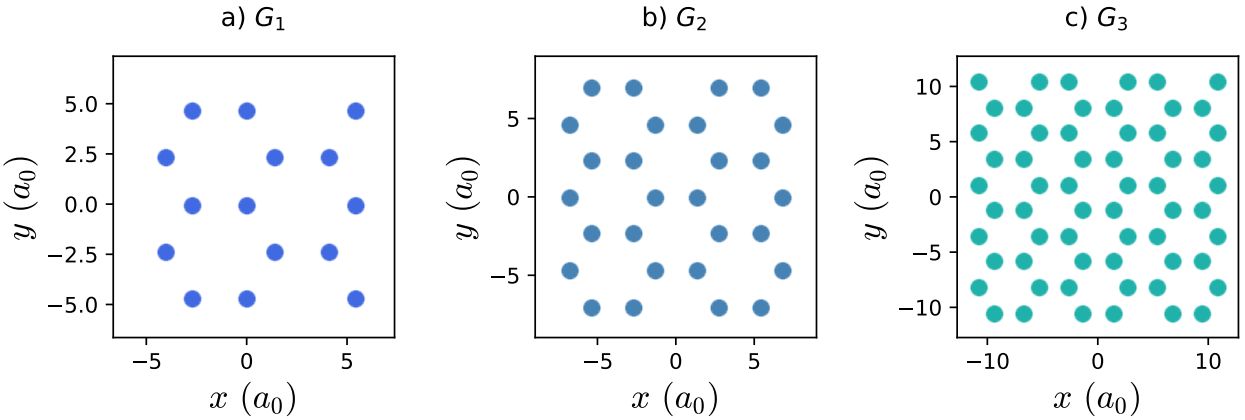


Figure 3.2: Crystal structures of the three Graphene flakes considered in this report. The G_1, G_2 and G_3 crystal lattices have totals of 15, 28 and 60 Carbon atoms, respectively. The neighbouring atoms in each lattice are separated by 1.42 angstrom, or approximately 2.68 Bohr radii (a_0).

3.2 Hückel Approximation

Slater basis functions (2.4) corresponding to the quantum numbers $l = 1$ and $m = 0$, and parameter $\lambda = 1.2$, were calculated for both Benzene and Graphene, over spatial domains of 10 and 20 Bohr radii respectively, and with 3D resolutions of (50x50x50) cells. With these, Hückel method Hamiltonians (2.7) for each system were also found, following their cyclic-planar structures. The following interaction matrix was used for Benzene, with the positions of the α and β parameters corresponding to a ring of Carbon atoms, each with two atomic neighbours:

$$\boldsymbol{F}^{(\boldsymbol{Benzene})} = \begin{pmatrix} \alpha & \beta & 0 & 0 & 0 & \beta \\ \beta & \alpha & \beta & 0 & 0 & 0 \\ 0 & \beta & \alpha & \beta & 0 & 0 \\ 0 & 0 & \beta & \alpha & \beta & 0 \\ 0 & 0 & 0 & \beta & \alpha & \beta \\ \beta & 0 & 0 & 0 & \beta & \alpha \end{pmatrix} \tag{3.1}$$

The interaction matrices for each Graphene crystal were constructed algorithmically by finding the nearest 2-3 neighbours to each Carbon atom, and placing β values at their associated matrix indices. For both Benzene and Graphene, the parameter values $\alpha = -0.6$ and $\beta = -0.1$ were used, corresponding to the $2p_z$ orbital energy and average interaction energy of Carbon atoms, respectively [12]. From these $\boldsymbol{F}$ and $\boldsymbol{S}$ interaction matrices, the general eigenvalue equations were then solved numerically, producing electron orbital eigenvalues and wavefunctions for each system. These were plotted, and then compared with known results [10].

3.3 Hartree-Fock Approximation

Gaussian basis functions (2.5) were first calculated for a range of Hydrogen systems, using the geometric basis set with $\alpha = 0.001$ and $\beta = 1.9$, and over spatial domains ranging from 10 to 40 Bohr radii. Basis functions for the orbital quantum numbers $l = 0$, $l = 1$ and $l = 2$ were generated with domain resolutions of (50x50x50), and the atomic Hartree-Fock $\boldsymbol{S}$, $\boldsymbol{T}$ and $\boldsymbol{V}$ matrices (2.15 - 2.17) were each calculated, combined, and solved numerically for the lowest-energy s, p and d states of Hydrogen. The resulting eigenvalues and wavefunctions were then plotted, and compared with expected results [14].

In further comparison, these steps were repeated using both the molecular Hartree-Fock equations (2.19 - 2.21), and a fully-numerical implementation of the Hartree-Fock matrix integrals (2.9 - 2.12), without the simplified analytic forms from (2.15) onwards. These resulting wavefunctions and eigenvalues were plotted and compared with the existing analytic, atomic results for s-shell, $l = 0$, $m = 0$ Hydrogen, to verify the accuracy of each method. Additionally, the calculation times for the analytic and numerical s-shell Hydrogen orbitals were also measured and compared, to determine the relative practicality of the numerical method.

Finally, additional Gaussian basis functions and molecular Hartree-Fock matrices were calculated for the atomic and molecular structures of Helium, Benzene and G_2 Graphene, over spatial domains from 10 to 40 Bohr radii. The resulting wavefunctions for Helium were calculated both with and without the addition of the atomic Hartree-Fock potential $G^{(HF)}$ (2.12), and these results were compared with known values [9] to evaluate the accuracy of both algorithms used. The eigenvalues and wavefunctions calculated for Benzene and Graphene were then both compared with their associated Hückel theory results.

4 Results and Discussion

4.1 Hydrogen Lowest-Energy States

4.1.1 Atomic and Molecular Hartree-Fock Methods

Figure 4.1 below shows orbital electron wavefunctions calculated for the six valid s, p, and d-type ($l = 0$, $l = 1$, and $l = 2$ respectively) lowest-energy Hydrogen states, found using the atomic Hartree-Fock approximation. Here, the shapes of each orbital closely match existing exact solutions for Hydrogen, found by decomposing the Schrödinger-equation wavefunction into radial and spherical-harmonic components [8][14]. The orbitals displayed are the real-valued components of each wavefunction, each of which occupies very distinct locations around the atomic centre, and has clear boundaries between positive and negative amplitude points.

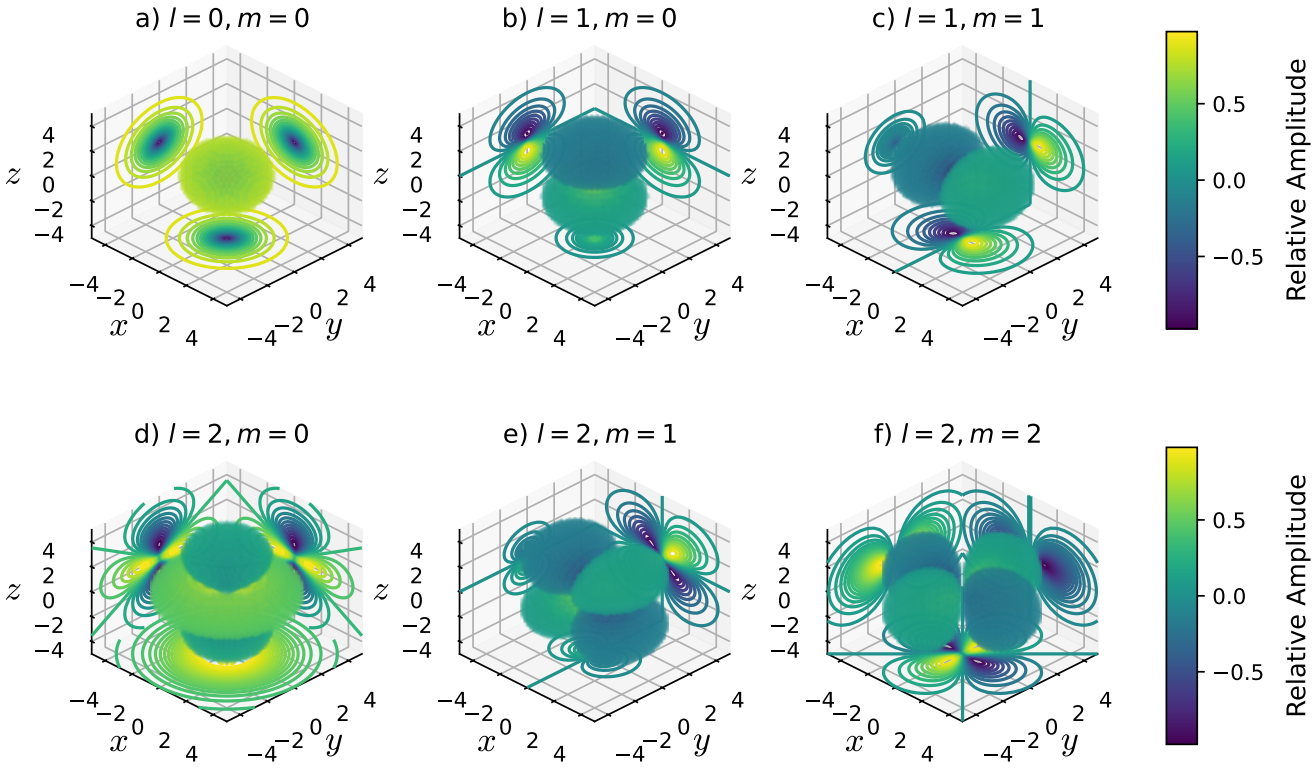


Figure 4.1: Orbital electron wavefunctions for the lowest-energy Hydrogen states with azimuthal quantum numbers from $l = 0$ to $l = 2$, calculated using the atomic Hartree-Fock approximation. These closely match known Hydrogen orbitals calculated analytically [14]. Displayed using a 3D scatterplot, and filtered to display real-valued wavefunction components greater than 10% relative amplitude. Calculated using a domain of (50x50x50) position cells.

Figure 4.2 below shows the first three electron orbital wavefunctions for s-type ($l = 0$, $m = 0$) Hydrogen, calculated with both the atomic and molecular forms of the Hartree-Fock approximation. Both methods produced identical wavefunctions with expected shapes, and so only the atomic result has been plotted. This confirms that the hydrogenic components of the molecular Hartree-Fock method ($\boldsymbol{S}$, $\boldsymbol{T}$, and $\boldsymbol{V}$) have been implemented correctly in the project script, and allows us to continue to more complex methods.

The first six ground-state eigenvalues found with these methods were also identical to one-another, and were $E_1 = -13.606\ eV$, $E_2 = -3.401\ eV$, $E_3 = -1.512\ eV$, $E_4 = -0.850\ eV$, $E_5 = -0.544\ eV$, and $E_6 = -0.313\ eV$. Of these, the first five closely match their exact Bohr model values of $-13.6/n^2\ eV$, however the sixth and further energies diverge quickly [11]. Therefore, the Hartree-Fock methods both give a reasonable approximation of the first few low-energy states, but become inaccurate quickly for those with higher energies.

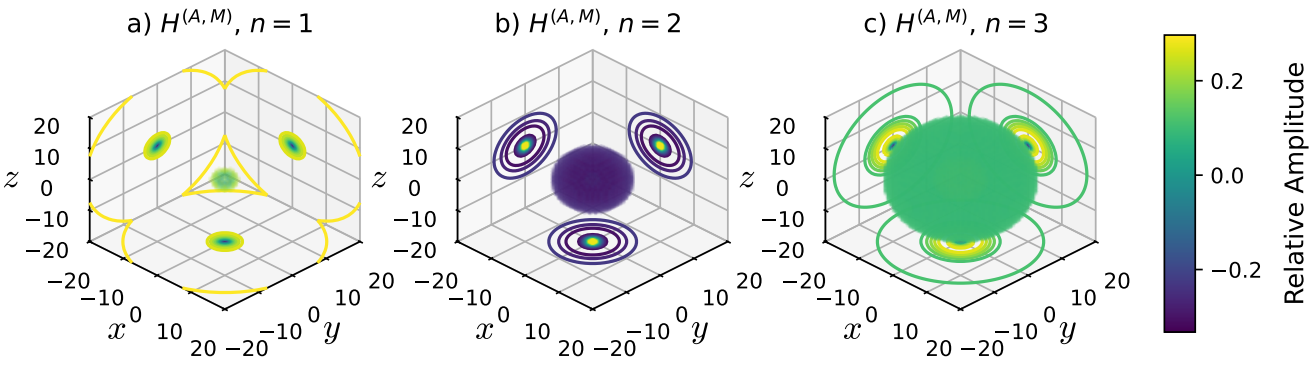


Figure 4.2: Orbital electron wavefunctions for $l = 0$, and $m = 0$ Hydrogen, calculated using the atomic Hartree-Fock approximation. The molecular form of the algorithm generated identical results, and has not been included. The relative radii of these orbitals can be seen increasing with a roughly n^2 relationship, matching the Bohr model of a Hydrogen atom [11]. Displayed using a 3D scatterplot, and filtered to display real-valued components greater than 10% relative amplitude.

Finally, Figure 4.3 below shows these same electron orbitals, but calculated with both the simplified atomic Hartree-Fock method, and a fully-numerical form of the general Hartree-Fock approximation. Here, the numerical wavefunctions match their analytic counterparts closely for the $n = 1$ and $n = 2$ states, but diverge significantly for $n = 3$ and further. The first six eigenvalue magnitudes found with this method were $|E_1| = -13.848\ eV$, $|E_2| = -3.477\ eV$, $|E_3| = |E_4| = -1.691\ eV$, and $|E_5| = |E_6| = 1.306\ eV$. These values are still close to their associated Bohr-model energies, but are less accurate than the analytical Hartree-Fock results. Additionally, they diverge massively by the fifth eigenvalue, and the magnitudes $|E_3|$ through to $|E_6|$ are doubly-degenerate due to the complex-valued solutions required from the inaccurate numerical model.

Notably, each of these results also used a (100x100x100) resolution domain to calculate the Gaussian basis functions and resulting matrices, as smaller domain sizes such as (50x50x50) resulted in significantly inaccurate values. Additionally, the computation times of these analytic and numerical methods were compared. On one device, the analytic method took an average of $0.41s$ to complete calculations over 10 iterations, while the numerical method took an average of $2.57s$. Clearly, for s-type Hydrogen, the analytical method is far superior to the numerical one. However, as the wavefunctions and eigenvalues of the two lowest-energy states were both reasonably accurate, the numerical method could be useful for more complex systems or alternative basis sets, where an analytic result does not exist.

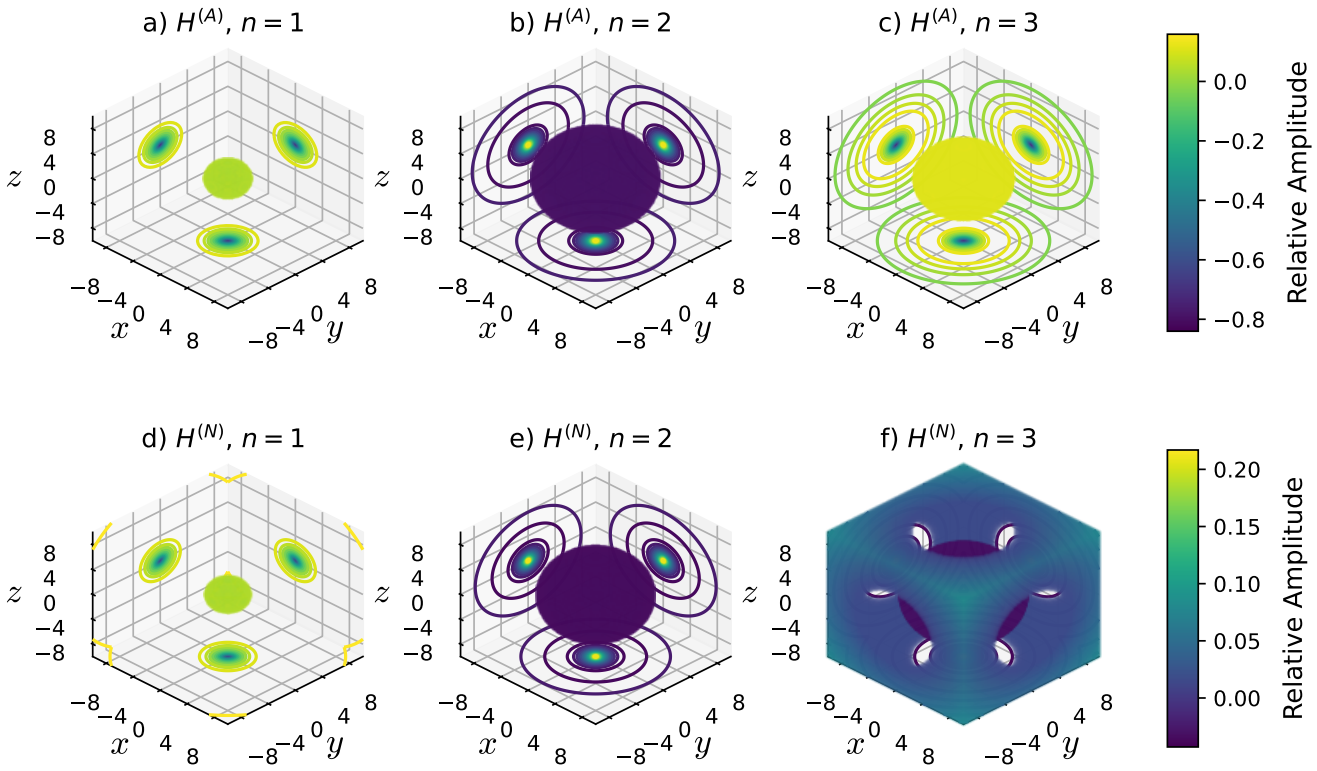


Figure 4.3: Orbital electron wavefunctions for $l = 0$, and $m = 0$ Hydrogen, calculated using analytical and numerical forms of the atomic Hartree-Fock approximation. The numerical wavefunctions $H^{(N)}$ match closely with the analytical results $H^{(A)}$ for $n = 1$ and $n = 2$, but diverge significantly for $n = 3$. Displayed using a 3D scatterplot, and filtered to display real-valued wavefunction components greater than 10% relative amplitude. Calculated using a domain of (100x100x100) position cells.

4.2 Helium S-Type Orbitals

4.2.1 Atomic Hartree-Fock Approximation

Figure 4.4 below shows the first three electron orbital wavefunctions for s-type ($l = 0$, $m = 0$) Helium, calculated with the atomic Hartree-Fock approximation, both with and without the Hartree-Fock potential $G^{(HF)}$. The wavefunctions calculated by ignoring electron-repulsion have similar shapes to those calculated for Hydrogen, and the first six eigenvalues calculated with these were $E_1 = -54.417\ eV$, $E_2 = -13.605\ eV$, $E_3 = -6.047\ eV$, $E_4 = -3.401\ eV$, $E_5 = -2.176\ eV$, and $E_6 = -1.509\ eV$. These closely match the exact values for one-electron He^+ Helium ions, whose energies correspond to the atom's binding energies [9].

For the Helium states calculated with the Hartree-Fock potential $G^{(HF)}$ however, the resulting wavefunction amplitudes were massively high, and incorrect. As the atomic Hartree-Fock algorithm used here was tested and proven accurate for single-electron Hydrogen states, the electron-repulsion component within the algorithm must be the point of error. By comparing individual Helium matrix elements with those in the "Complex atomic exercises" document, linked to the "Quantum Chemistry Project" guide [12], it was determined that all matrix values were correct except for those from the density matrix D_{kl}. Unfortunately, the source of this error could not be found, despite attempts at scaling and normalising these matrix values.

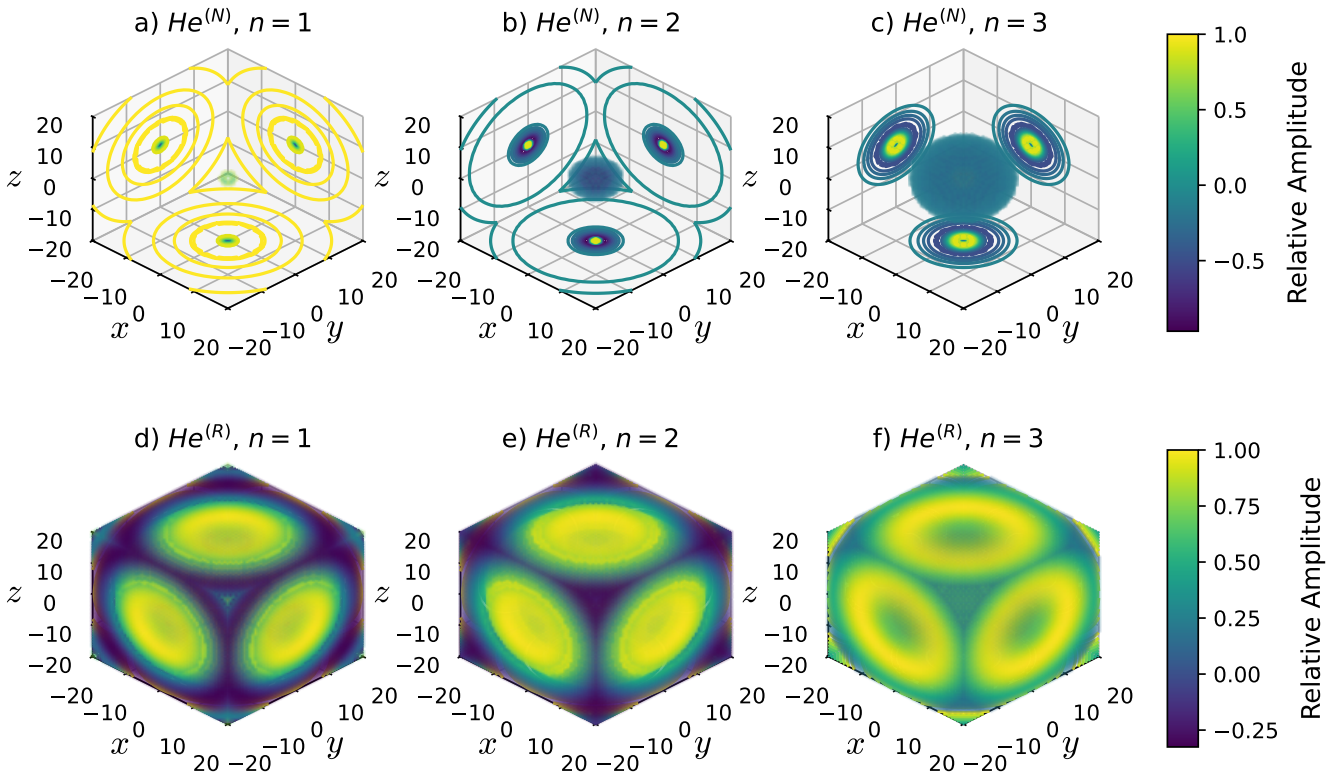


Figure 4.4: Orbital electron wavefunctions for $l = 0$, and $m = 0$ Helium, calculated using the atomic Hartree-Fock approximation, and displayed using a 3D scatterplot. The Hartree-Fock potential $G^{(HF)}$ was ignored when calculating subplots a) through c), but included for subplots d) through f). The wavefunctions calculated without electron repulsion match one-electron He^{+} Helium states [9], but those calculated with repulsion have incorrect, massively high amplitudes. Clearly, there is a significant error in wavefunction amplitudes produced by the electron-repulsion algorithm within this project.

4.3 Benzene Orbitals

4.3.1 Hückel Approximation

Figure 4.5 below shows orbital electron wavefunctions calculated for the six lowest-energy states of $l = 1$, $m = 0$ Benzene, found using the Hückel approximation. Here, the shapes of each orbital closely match existing solutions for Benzene, found both with industry-standard Hartree-Fock and density-functional calculations [13], and with an analytic Hückel approximation [3], confirming a correct algorithmic implementation. Here, the highest-energy state shows distinct, separate orbital sites around each Carbon atom in the molecule, while the lowest-energy state has electron probabilities distributed equally across the almost-circular, constant-z surfaces of the Benzene ring.

The six eigenvalues calculated corresponding to these states were $E_1 = -21.769\ eV$, $E_2 = E_3 = -19.048\ eV$, $E_4 = E_5 = -13.606\ eV$, and $E_6 = -10.885\ eV$, equivalent to $\alpha \pm \beta$ or $\alpha \pm 2\beta$ for the values of $\alpha = -0.6$ and $\beta = -0.1$ chosen, in these units [12]. These values do not appear to match any exact solutions, but as both E_2 and E_3, and E_4 and E_5 are doubly-degenerate, these values may be linear combinations of an exact solution [12]. Notably, E_4 and E_5 are identical to the Hartree-Fock estimate for Hydrogen's ground-state energy. It is unclear whether this indicates some underlying behaviour, or is just coincidence.

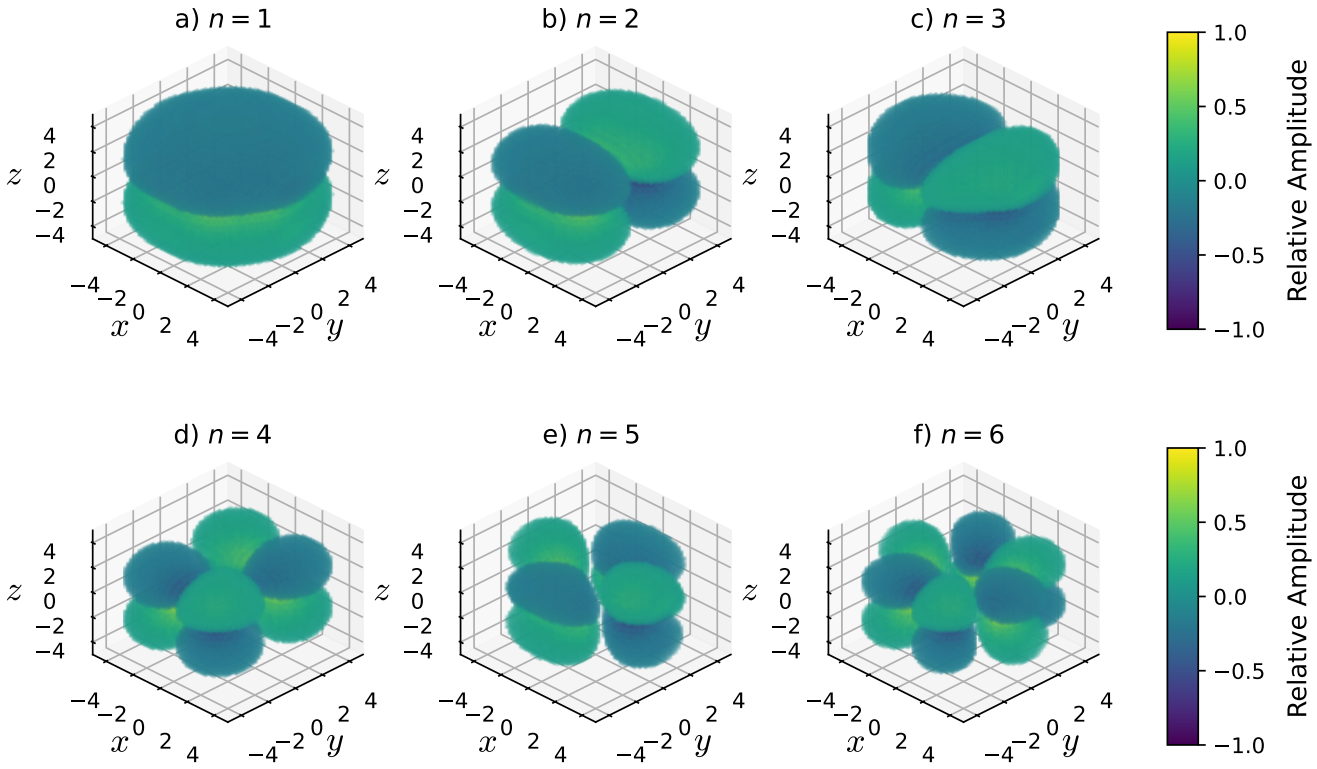


Figure 4.5: Orbital electron wavefunctions for $l = 1$, and $m = 0$ Benzene, calculated using the Hückel approximation. These orbital shapes closely match known Benzene orbitals calculated through Hartree-Fock and density-functional theory [13]. Displayed using a 3D scatterplot, and filtered to display real-valued wavefunction components greater than 10% relative amplitude.

4.3.2 Hartree-Fock Molecular Approximation

Figure 4.6 below shows orbital electron wavefunctions calculated for $l = 1$, $m = 0$ Benzene, using the molecular Hartree-Fock approximation. Although there is some structure to the calculated orbitals, in comparison with the Hückel method results, these are obviously incorrect. This appears to be due to the same error found earlier in the Hartree-Fock potential, $G^{(HF)}$.

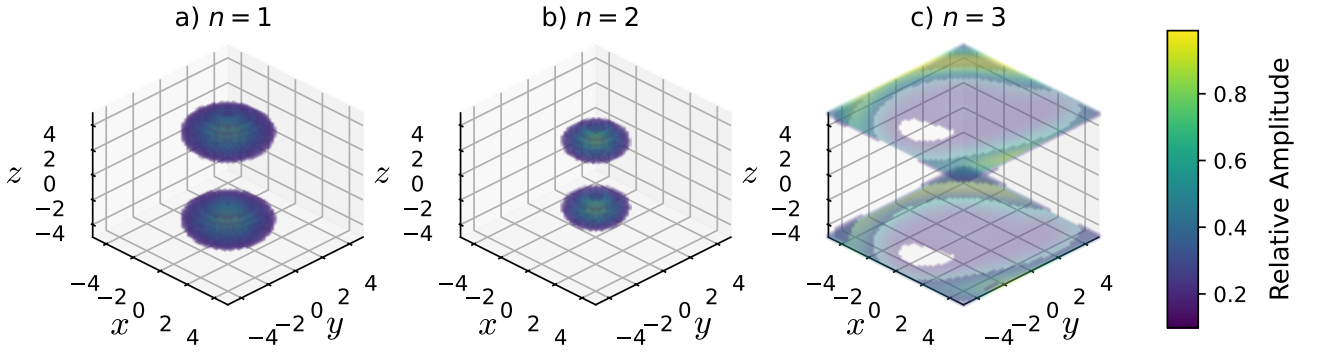


Figure 4.6: Orbital electron wavefunctions for $l = 1$, and $m = 0$ Benzene, calculated using the molecular Hartree-Fock approximation, and displayed using a 3D scatterplot. Although there is some structure to the calculated orbitals, they are obviously incorrect. This appears to be due to the same error identified in the calculated Hartree-Fock potential $G^{(HF)}$, as seen in Figure 4.4.

4.4 Graphene Orbitals

4.4.1 Hückel Approximation

Figures 4.7 and 4.8 below show orbital electron wavefunctions calculated for evenly-spaced samples of six and three, $l = 1$, $m = 0$, G_2 Graphene states, found using the Hückel approximation. Here, the shapes of the higher-energy orbitals closely match expected behaviour for Graphene, as a periodic crystal lattice [10]. The highest-energy state plotted shows distinct, separate orbital sites around each Benzene ring within the Graphene molecule. Notably, as with the molecular wavefunctions for Benzene, the lowest-energy state has electron probabilities distributed almost-equally across the entire surface of the molecule. Additionally, some of the resulting states with energies in-between these, $n = 13$ and $n = 17$, have interesting one-dimensional periodicity, corresponding to Benzene ring positions with the molecule.

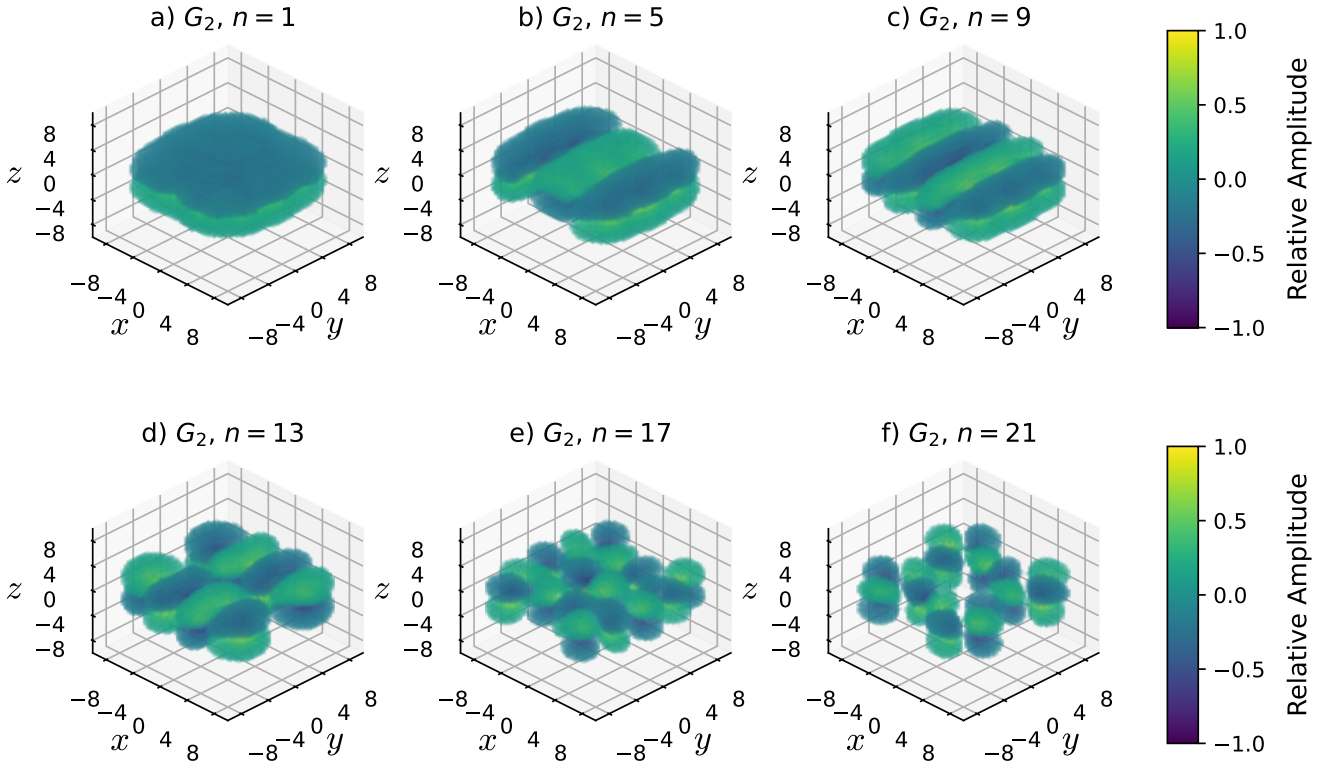


Figure 4.7: Orbital electron wavefunctions for $l = 1$, and $m = 0$, G_2 Graphene, calculated using the Hückel approximation. Starting at $n = 1$ and ending with $n = 21$, every fourth energy state is displayed using a 3D scatterplot of the wavefunction, to showcase a greater variety of Graphene orbitals. These match the periodic structure expected from Graphene's 2D crystal lattice, with the higher-energy orbitals appearing periodic over the Graphene crystal's unit cell [10]. Filtered to display real-valued wavefunction components greater than 10% relative amplitude.

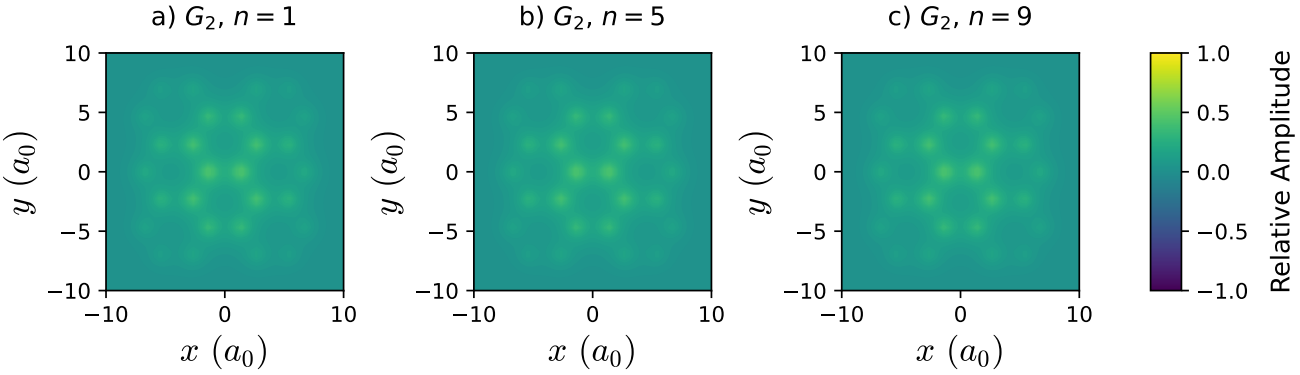


Figure 4.8: Orbital electron wavefunctions for $l = 1$, and $m = 0$, G_2 Graphene, calculated using the Hückel approximation. Starting at $n = 1$ and ending with $n = 9$, every fourth energy state is displayed using a $z = 0$, 2D contour of the 3D wavefunction, to showcase a greater variety of Graphene orbitals. With this particular 2D slice, the atomic structure of the Graphene crystal is quite visible.

Finally, Figure 4.9 below shows the highest-energy ($n = 15$, $n = 28$, and $n = 60$) states for the three Graphene molecules G_1, G_2, and G_3, respectively. These show that the periodic structure of the orbital wavefunctions calculated is consistent across molecule sizes, but displays slightly different amplitudes and shapes at the boundaries of each molecule. For a much larger Graphene molecule, it would then be sufficient to model the electron wavefunctions and probabilities periodically, using a Fourier-transformed reciprocal lattice of the spatial lattice used here [10]. This could be interesting further study, as the reciprocal lattice can be described with less information than its spatial counterpart, and could be simulated with much higher resolutions (albeit without this form of the Hückel approximation).

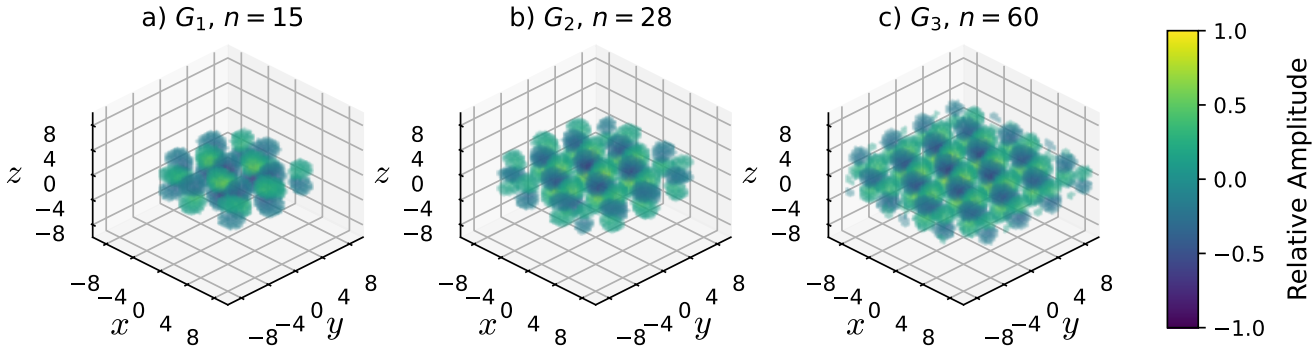


Figure 4.9: The highest-energy orbital electron wavefunctions of G_1, G_2, and G_3 Graphene, calculated using the Hückel approximation. These each match the periodic structure expected from Graphene's 2D crystal lattice [10]. Filtered to display real-valued wavefunction components greater than 10% relative amplitude.

4.4.2 Hartree-Fock Molecular Approximation

The G_2 electron orbitals were also calculated using the molecular Hartree-Fock approximation, but had no visible structure due to the electron-repulsion error discussed earlier, and thus are not plotted. These calculations are more good candidates for further study, as the molecular Hartree-Fock method considered more complicated electron-electron energy contributions [12], and could display significant deviations from the Hückel method results, particularly at the molecular boundaries.

5 Conclusion

Electron orbital behaviour within both atomic and cyclic-molecular systems was investigated using the Hückel and Hartree-Fock approximations. When compared with experimental results [13][10], the Hückel method produced accurate electronic wavefunctions for both Benzene molecules and three differently-sized Graphene crystals, but corresponding eigenvalues were not confirmed to be accurate. The Hartree-Fock algorithm [6][12] produced accurate s-type Hydrogen energies and wavefunctions in both analytical and numerical forms, but required more computationally-expensive domain resolutions for accurate fully-numerical orbitals, suggesting utility only in systems without analytic alternatives. In comparing s-type Helium orbitals, an error was found in the algorithm used for the Hartree-Fock potential $G^{(HF)}$, and therefore, molecular Hartree-Fock results for both Benzene and Graphene could not be properly compared to those from the Hückel method.

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